OpenPipeline Migration Guide: Part 2

> **Series:** OPMIG | **Notebook:** 2 of 9 | **Created:** December 2025 | **Last Updated:** 01/28/2026

Architecture & Key Concepts

Learning Objectives

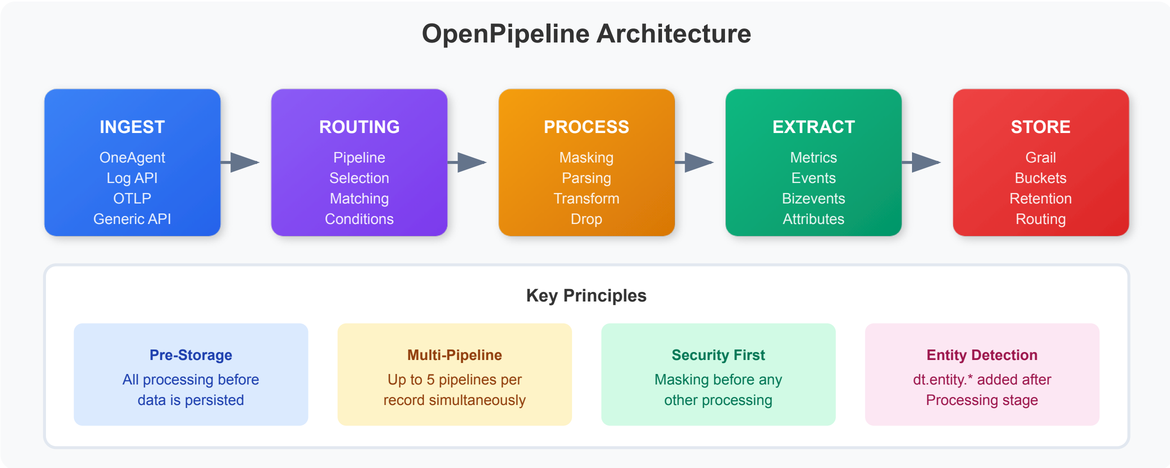
By the end of this notebook, you will:

- ✅ Understand the complete OpenPipeline data flow architecture
- ✅ Learn detailed processing stage execution order
- ✅ Master DPL (Dynatrace Pattern Language) fundamentals
- ✅ **Know ALL OpenPipeline limits in comprehensive detail**
- ✅ Understand processor types and capabilities
- ✅ **Learn entity field availability timeline**
- ✅ Explore your pipeline configuration with DQL

Data Flow Architecture

Understanding how data flows through OpenPipeline is essential for designing effective pipelines.

Complete Data Flow



Key Principles

1. **Pre-storage Processing**: All transformations happen BEFORE data is written to Grail
2. **Order Matters**: Processors execute in defined order within each stage
3. **Multi-pipeline Support**: A single record can be processed by up to 5 pipelines
4. **Entity Detection**: Entity fields (`dt.entity.*`) are added AFTER the Processing stage
5. **Immutable Storage**: Once stored, data cannot be modified (mask early!)

Processing Stages in Detail

Stage 1: Routing

The routing stage determines which pipeline(s) process each incoming record.

Aspect	Description
Purpose	Match incoming data to appropriate pipelines
Timing	First stage - before any processing
Configuration	Dynamic routing rules with matching conditions
Fallback	Unmatched data goes to default pipeline

Matching Condition Examples:

```
k8s.namespace.name == "production"
log.source == "nginx"
contains(content, "payment")
dt.openpipeline.source == "oneagent"
```

Stage 2: Processing

The processing stage transforms, enriches, and filters data.

Sub-stages (in order):

1. **Masking** - Redact sensitive data (applied FIRST for security)
2. **Filtering** - Drop unwanted records
3. **Processing** - Parse, transform, enrich

Processor Type	Purpose	Example
DQL	Transform with DQL commands	`fieldsAdd`, `fieldsRemove`, `parse`
Drop	Remove matching records	Drop debug logs
Technology Parser	Apply built-in parsers	Apache, JSON, syslog

Stage 3: Extraction

The extraction stage creates derived data from processed records.

Extraction Type	Output	Use Case
Value Metric	Numeric metric with dimensions	Response times, counts
Counter Metric	Incrementing counter	Request counts, errors
Event	Platform event	Custom alerts, notifications
Business Event	Business analytics event	Transactions, user actions

Stage 4: Storage

The storage stage routes data to Grail buckets.

Aspect	Description
Bucket Routing	Direct data to specific buckets
Retention	Different retention periods per bucket
Cost Control	Route high-volume data to shorter retention

Pipeline Types

OpenPipeline supports three types of pipelines:

1. Default Pipeline

- **Purpose**: Handles data that doesn't match any custom pipeline
- **Configuration**: Minimal processing, default bucket
- **Modification**: Can add processors but cannot delete

2. Custom Pipelines

- **Purpose**: Targeted processing for specific data sources or patterns
- **Configuration**: Full control over all processing stages
- **Routing**: Requires matching conditions in dynamic routing

3. Built-in Pipelines

- **Purpose**: Pre-configured for common technologies
- **Examples**: Kubernetes logs, AWS logs, Azure logs
- **Modification**: Can extend but not fundamentally change

> 💡 **Best Practice:** Create focused pipelines for specific use cases rather than one large pipeline with complex conditionals.

Processor Types

DQL Processor

The DQL processor uses DQL commands to transform data. Available commands:

Command	Purpose	Example
`fieldsAdd`	Add new fields	`fieldsAdd environment = "prod"`
`fieldsRemove`	Remove fields	`fieldsRemove sensitive_field`
`fieldsRename`	Rename fields	`fieldsRename old_name = new_name`
`parse`	Extract with DPL patterns	`parse content, "LD:prefix INT:count"`

fieldsAdd Examples:

```
// Static value
| fieldsAdd environment = "production"

// Conditional value
| fieldsAdd severity = if(loglevel == "ERROR", "critical", else: "normal")

// Computed value
| fieldsAdd message_length = stringLength(content)

// From existing field
| fieldsAdd short_host = substring(host.name, 0, 10)
```

Drop Processor

Removes records matching specified conditions:

```
Matching Condition: loglevel == "DEBUG"
Result: All DEBUG logs are dropped before storage
```

Technology Processors

Built-in parsers for common log formats:

Technology	Parsed Fields
Apache	client_ip, method, path, status, bytes
Nginx	Similar to Apache with nginx-specific fields
JSON	Flattens JSON structure to fields
Syslog	facility, severity, hostname, message

Dynatrace Pattern Language (DPL)

DPL is a powerful pattern matching language used in the `parse` command.

Core Matchers

Matcher	Description	Example Match
`INT`	Integer number	`42`, `-17`
`LONG`	Long integer	`1234567890123`
`DOUBLE`	Decimal number	`3.14`, `-0.5`
`IPADDR`	IPv4 or IPv6 address	`192.168.1.1`, `::1`
`IPV4ADDR`	IPv4 address only	`10.0.0.1`
`IPV6ADDR`	IPv6 address only	`2001:db8::1`
`TIMESTAMP`	Timestamp with format	`2024-01-15T10:30:00Z`
`LD`	Line data (to delimiter)	Any text until delimiter
`DATA`	Any data (greedy)	Consumes remaining text
`SPACE`	Whitespace	Spaces, tabs
`NSPACE`	Non-whitespace	Any non-space chars
`WORD`	Word characters	`hello`, `user123`
`JSON`	JSON structure	`{"key": "value"}`
`EOL`	End of line	Line terminator

Pattern Syntax

Element	Syntax	Description
Export to field	`MATCHER:fieldname`	Extract and name the field
Match only	`MATCHER`	Match but don't extract
Optional	`MATCHER?`	Matcher is optional
Literal	`'exact text'`	Match literal string
Alternatives	`('opt1'|'opt2')`	Match either option
Quantifier	`MATCHER{2,5}`	Match 2-5 times

DPL Examples

```
// Extract IP and port from log
| parse content, "IPADDR:client_ip ':' INT:port"

// Extract user ID with flexible prefix
| parse content, "('user='|'userId='|'user_id=')LD:user_id"

// Parse Apache-style log
| parse content, "IPADDR:client_ip SPACE '-' SPACE LD:user SPACE '['
LD:timestamp ']'"

// Extract JSON payload
| parse content, "LD JSON:payload"

// Parse with optional port
| parse content, "IPADDR:ip (':' INT:port)?"

// Extract error code
| parse content, "'error_code=' INT:error_code"
```

Timestamp Format Patterns

Symbol	Meaning	Example
`yyyy`	Year (4 digits)	2024
`MM`	Month (01-12)	01
`dd`	Day (01-31)	15
`HH`	Hour 24h (00-23)	14
`mm`	Minute (00-59)	30
`ss`	Second (00-59)	45
`SSS`	Milliseconds	123

Complete OpenPipeline Limits Reference

This comprehensive reference contains ALL OpenPipeline limits you need to know for planning your migration.

Data Size & Volume Limits

Limit	Value	Behavior When Exceeded
Max record size (after processing)	16 MB	Record is **dropped**, logged in audit
Max record size (before processing)	1 MB	Ingestion rejected with 413 error
Working memory per record	16 MB	Processing fails, record dropped
Log attribute size	32 KB	Attribute value **truncated**
Max field name length	255 characters	Field creation fails
Max string field length	4 KB	Content truncated
Max array size	1000 elements	Array truncated
Max nesting depth (JSON)	10 levels	Deeper levels flattened

Processing Limits

Limit	Value	Impact
Max pipelines per record	5	Up to 5 pipelines can process one record
Max processors per pipeline	50	Cannot add more processors to pipeline
Max DQL commands per processor	10 commands	Split complex logic into multiple processors
Max parse operations per processor	100 patterns	Create additional parse processors
Max fields per record	1000 fields	Additional fields ignored
Processing timeout per record	30 seconds	Record dropped if exceeded
Max processor name length	100 characters	Validation error

Timestamp Constraints

Data Type	Accepted Range	Records Outside Range
Logs	24 hours past to 10 minutes future	**Dropped**
Spans	2 hours past	**Dropped**
Events	24 hours past to 10 minutes future	**Dropped**
Business Events	24 hours past to 10 minutes future	**Dropped**
Metrics	1 hour past to 1 minute future	**Dropped**

> ⚠️ **Critical:** Historical data imports require workarounds. Contact Dynatrace support for backfilling options.

Pipeline & Routing Limits

Limit	Value	Scope
Max custom pipelines	100 pipelines	Per configuration scope (logs, spans, etc.)
Max dynamic routes	100 routes	Per configuration scope
Max conditions per route	10 conditions	Combine with AND/OR operators
Max pipeline name length	100 characters	Validation error
Max route name length	100 characters	Validation error

Extraction Limits

Limit	Value	Notes
Max metric extractions per pipeline	10	Value + counter metrics combined
Max event extractions per pipeline	5	All event types combined
Max bizevent extractions per pipeline	3	Business events only
Max dimensions per metric	10 dimensions	Keep cardinality low
Metric key length	250 characters	Validation error
Event type name length	100 characters	Validation error

DQL & DPL Limits

Limit	Value	Context
Max DQL query length	64 KB	In processor definition
Max DPL pattern length	4 KB	Per parse pattern
Max captured groups per parse	100 groups	Use multiple parse operations
Max alternatives in pattern	50 alternatives	`(opt1|opt2|...|opt50)`

Bucket & Storage Limits

Limit	Value	Notes
Max custom buckets	50 buckets	Per environment
Min retention period	1 day	Per bucket
Max retention period	2555 days (~7 years)	Per bucket
Bucket name length	100 characters	Alphanumeric + underscore

Rate Limits

Endpoint	Limit	Per
Log Ingest API	500 requests/min	Per token
OTLP Endpoint	1000 requests/min	Per token
Config API	100 requests/min	Per token

Field & Matching Restrictions

Read-Only Fields (Cannot be modified):

```
dt.ingest.*          - Ingestion metadata
dt.openpipeline.*    - Pipeline processing metadata
dt.retain.*          - Retention information
dt.system.*          - System metadata (bucket, etc.)
timestamp            - Record timestamp (can parse new, not modify original)
```

Reserved Field Prefixes (Cannot create):

```
dt.*                  - Reserved for Dynatrace
dynatrace.*           - Reserved for Dynatrace
```

Entity Fields (Available ONLY after Processing stage):

```
dt.entity.service
dt.entity.host
dt.entity.process_group
dt.entity.process_group_instance
dt.entity.kubernetes_cluster
dt.entity.cloud_application
dt.entity.cloud_application_namespace
```

> 💡 **Design Pattern:** Entity fields are added by Dynatrace AFTER the Processing stage. You cannot:
> - Use them in routing conditions
> - Use them in matching conditions during processing
> - Modify them with fieldsAdd/fieldsRename
>
> You CAN use them in:
> - Extraction stage (metrics, events)
> - Storage stage (bucket routing based on entity)

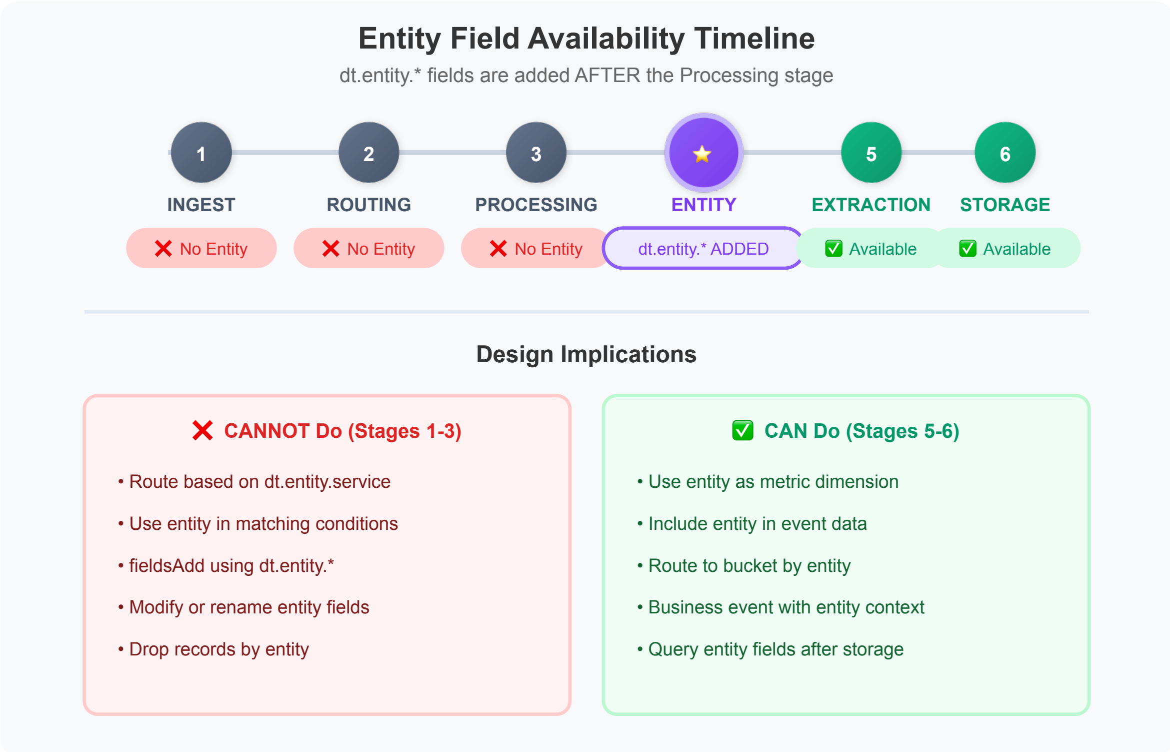
Workarounds for Common Limit Issues

Problem	Workaround
>50 processors needed	Split into multiple pipelines, use multi-pipeline routing
>10 DQL commands	Break into multiple processors (order matters!)
>100 parse patterns	Use multiple parse processors in sequence
>16MB after processing	Drop unnecessary fields, reduce field sizes
>5 pipelines needed	Consolidate processing logic, use conditional processors
>10 metric dimensions	Reduce cardinality, use separate metrics

Entity Field Availability Timeline

Understanding WHEN entity fields become available is critical for pipeline design.

Processing Stage Timeline



Practical Examples

❌ WRONG - Cannot route based on entity

```
Dynamic Route:
  Condition: dt.entity.service == "SERVICE-123"  ❌ FAILS

Why: Entity fields not available during routing
```

❌ WRONG - Cannot use entity in processing

```
DQL Processor:
  fieldsAdd service_name = dt.entity.service  ❌ FAILS

Why: Entity fields not available during processing
```

✅ CORRECT - Use entity in metric extraction

```
Metric Extraction:
  Key: log.request.duration
  Value: duration_ms
  Dimensions: dt.entity.service, dt.entity.host  ✅ WORKS

Why: Entity fields available in extraction stage
```

✅ CORRECT - Use entity for bucket routing

```
Bucket Routing (in Storage stage):
  Condition: dt.entity.service == "CRITICAL-SERVICE"
  Bucket: critical_logs  ✅ WORKS

Why: Entity fields available in storage stage
```

Design Patterns

Pattern 1: Route by source, extract metrics by entity

```
Routing: log.source == "payment-service"  (uses source field)
Processing: Parse payment data
Extraction: Create metric with dt.entity.service dimension  ✅
```

Pattern 2: Enrich with custom field, then use entity

```
Processing: fieldsAdd app_tier = "frontend"  (custom field)
Extraction: Dimensions: app_tier, dt.entity.service  ✅
```

Pattern 3: Cannot mix entity with routing

```
❌ Route based on entity → Use workaround:
   - Route based on source/content patterns instead
   - Use service name string (not entity ID) if available
```

Key Fields & Metadata

OpenPipeline-Specific Fields

Field	Description	Example Value
`dt.openpipeline.source`	Data source identifier	`oneagent`, `generic`, `otlp`
`dt.openpipeline.pipelines`	Pipeline(s) that processed record	`["custom-pipeline-1"]`
`dt.system.bucket`	Grail storage bucket	`default_logs`, `custom_logs`

Log Fields

Field	Description
`timestamp`	Log timestamp (primary time field)
`content`	Log message content
`loglevel`	Log level (ERROR, WARN, INFO, DEBUG)
`status`	Status string (alternative to loglevel)
`log.source`	Log source identifier
`log.iostream`	Stream type (stdout, stderr)

Entity Context Fields

Field	Description
`dt.entity.host`	Host entity ID
`dt.entity.process_group`	Process group entity ID
`dt.entity.process_group_instance`	Process instance entity ID
`dt.entity.service`	Service entity ID
`host.name`	Host name (string)
`process.executable.name`	Process executable name

> ⚠️ **Important:** Entity fields (`dt.entity.*`) are added AFTER the Processing stage. You cannot use them in routing or processing conditions.

Exploring Your Pipeline Configuration

Use these queries to understand how your environment is configured.

```
// View data sources currently sending to OpenPipeline
// Shows the distribution of data by ingestion source
fetch logs, from: now() - 24h
| summarize {record_count = count()}, by: {dt.openpipeline.source}
| sort record_count desc
```

```
// Analyze which pipelines are processing your logs
// Helps verify routing is working correctly
fetch logs, from: now() - 24h
| filter isNotNull(dt.openpipeline.pipelines)
| summarize {record_count = count()}, by: {dt.openpipeline.pipelines}
| sort record_count desc
```

```
// Check bucket distribution for stored logs
// Verify data is routing to expected buckets
fetch logs, from: now() - 24h
| summarize {record_count = count()}, by: {dt.system.bucket}
| sort record_count desc
```

```
// Analyze pipeline processing by source and pipeline
// Shows the relationship between sources and pipelines
fetch logs, from: now() - 24h
| filter isNotNull(dt.openpipeline.pipelines)
| summarize {record_count = count()}, by: {dt.openpipeline.source,
dt.openpipeline.pipelines}
| sort record_count desc
| limit 25
```

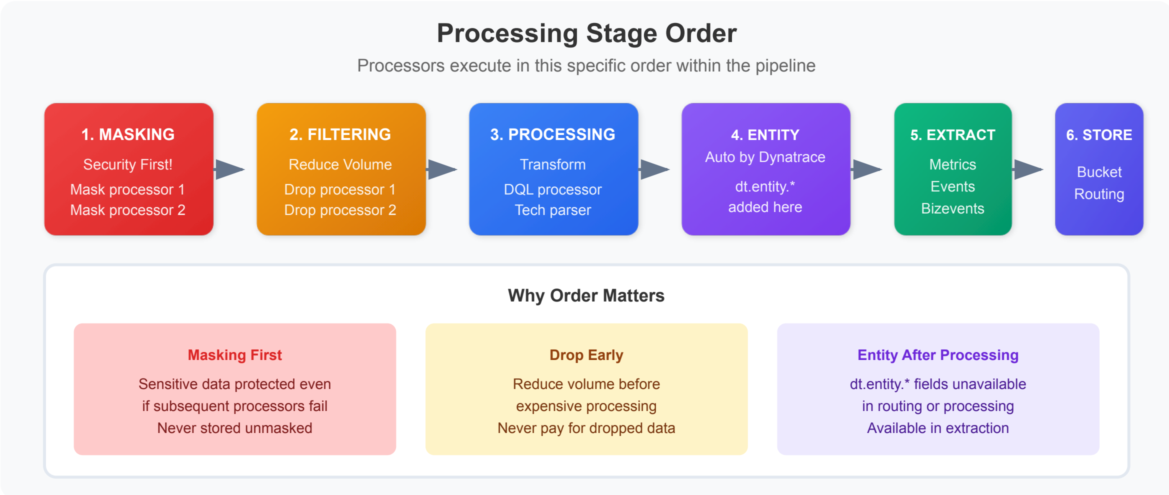
```
// Check for span processing through OpenPipeline
// Spans also flow through OpenPipeline
fetch spans, from: now() - 24h
| summarize {span_count = count()}, by: {dt.openpipeline.source}
| sort span_count desc
```

```
// View pipeline processing over time
// Helps identify volume patterns and processing trends
fetch logs, from: now() - 24h
| filter isNotNull(dt.openpipeline.pipelines)
| makeTimeseries {record_count = count()}, by: {dt.openpipeline.pipelines},
interval: 1h
```

```
// Identify logs going to default pipeline (may need custom routing)
// High volume in default may indicate missing routing rules
fetch logs, from: now() - 24h
| filter isNull(dt.openpipeline.pipelines) OR dt.openpipeline.pipelines == "
[]"
| summarize {unrouted_count = count()}, by: {log.source}
| sort unrouted_count desc
| limit 20
```

Understanding Processing Order

The order of processing within a pipeline is critical:



> 💡 **Tip:** Masking is applied FIRST so sensitive data is protected even if subsequent processors fail.

Summary: Key Architecture Concepts

Concept	Key Points
Data Flow	Ingest → Route → Process → Extract → Store
Pre-storage	All processing happens before data is persisted
Routing	Dynamic routes match data to pipelines
Processing Order	Masking → Filtering → Processing
Entity Detection	Happens AFTER processing, BEFORE extraction
Multi-pipeline	One record can be processed by up to 5 pipelines
DPL	Powerful pattern language for parsing
Buckets	Control retention and cost at storage stage

Next Steps

Now that you understand OpenPipeline architecture, continue with:

Notebook	Focus Area
OPMIG-03	Migration Assessment & Planning
OPMIG-04	Pipeline Configuration Fundamentals
OPMIG-05	Routing & Bucket Management
OPMIG-06	Processing, Parsing & Transformation

References

- OpenPipeline Data Flow
- OpenPipeline Processing
- Dynatrace Pattern Language
- DPL Architect Tool

Last Updated: December 12, 2025

This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.